R99/A | Locked full18 results

Same unique VAL checkpoint on VAL/TEST/LIB; native/FP64 ind; historical NSS weights disclosed.

epoch 39 | update 56160 | seed2 | fresh initialization

Data: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

Split / Problem	Top1 (%)	SBS Top1 (%)	mean_cost	SBS cost	Oracle cost	vs_SBS (%)	vs_Oracle (%)	actual regret (%)
VAL ALL	71.0938	66.2037	11.416896	11.427978	11.371692	-0.1924	+0.3952	0.4052
TEST ALL	71.3321	66.1802	11.373828	11.385299	11.329932	-0.1986	+0.3845	0.3930

Dataset	n	NSS Top1 (%)	R99 Top1 (%)	NSS cost	R99 cost
TSP TEST	4500	45.888889	47.311111	7.970400719	7.970608504
CVRP TEST	4500	45.977778	47.955556	18.004802368	18.010963113
TSPLIB	49	20.408163	26.530612	8.217907074	8.207570426
CVRPLIB	100	70.000000	64.000000	67.101111717	67.240244207

NSS: R79 weights trained on historical70114 TRAIN, selected using historical VAL; not retrained on new90k.

NSS TEST predictions were freshly computed on these4500 instances; old3895 TEST metrics are not reused.

SBS is fitted on the new81000 TRAIN only. Oracle is the per-instance legal-pool minimum.

ALL is the equal-problem mean; mixed raw cost is descriptive. TEST informed prior research.

Method selection and true winner use separate colors on the same axes; Top1 is per-instance accuracy.

Selection head: original_score_mlp_maximin | objective: original R77 CE .35 / comparison .35 / risk .02 / R-Drop .35

R99/A | VAL: ALL and18 tasks

TRAIN-fitted SBS and unchanged native/FP64 costs; no per-task checkpoint assembly.

Split / Problem	Top1 (%)	SBS Top1 (%)	mean_cost	SBS cost	Oracle cost	vs_SBS (%)	vs_Oracle (%)	actual regret (%)	TRAIN SBS
ALL	71.0938	66.2037	11.416896	11.427978	11.371692	-0.1924	+0.3952	0.4052	per-problem TRAIN SBS
TSP	47.1333	33.1111	8.025246	8.045354	7.990533	-0.2499	+0.4344	0.4232	T2T
CVRP	46.8000	18.7111	18.160692	18.269377	18.064087	-0.5949	+0.5348	0.7108	RELD_CVRP
ATSP	77.1111	47.4889	1.638478	1.676905	1.632471	-2.2915	+0.3680	0.3649	UNICO_MatPOENet
OVRP	41.1778	35.7556	7.461782	7.469579	7.415777	-0.1044	+0.6204	0.6339	MoSES_RF
VRPB	81.0222	80.8444	9.356391	9.354876	9.318750	+0.0162	+0.4039	0.4067	reld_mt
VRPL	37.9333	33.0889	12.431506	12.437959	12.383444	-0.0519	+0.3881	0.3960	MoSES_RF
VRPTW	80.0222	79.9556	18.818687	18.819431	18.766139	-0.0040	+0.2800	0.2958	reld_mt
OVRPTW	83.8444	83.7778	10.143000	10.143730	10.116859	-0.0072	+0.2584	0.2557	reld_mt
OVRPB	88.4444	88.5111	6.357919	6.357254	6.341078	+0.0105	+0.2656	0.2785	reld_mt
OVRPL	40.9111	35.9111	7.437541	7.445709	7.390416	-0.1097	+0.6377	0.6536	MoSES_RF
VRPBL	78.7778	78.5111	9.368539	9.371523	9.328366	-0.0318	+0.4306	0.4325	reld_mt
VRPBTW	78.5556	78.4222	19.947207	19.950430	19.851084	-0.0162	+0.4842	0.4629	reld_mt
VRPLTW	78.8889	78.7556	18.765126	18.765871	18.707843	-0.0040	+0.3062	0.3223	reld_mt
OVRPBL	89.2889	89.2000	6.369639	6.370093	6.352824	-0.0071	+0.2647	0.2764	reld_mt
OVRPBTW	83.5556	83.4444	10.660566	10.661098	10.623245	-0.0050	+0.3513	0.3249	reld_mt
OVRPLTW	83.5556	83.4889	10.157040	10.157423	10.129818	-0.0038	+0.2687	0.2723	reld_mt
VRPBLTW	79.5111	79.4000	19.794133	19.797044	19.706420	-0.0147	+0.4451	0.4332	reld_mt
OVRPBLTW	83.1556	83.2889	10.610629	10.609955	10.571302	+0.0063	+0.3720	0.3493	reld_mt

vs_SBS=100*(mean_cost/SBS_cost-1); vs_Oracle=100*(mean_cost/Oracle_cost-1).

actual regret=mean_i[100*(selected_i-oracle_i)/oracle_i]; not a ratio of means.

TRAIN81000 / VAL4500 / TEST4500 per problem. All18 problems have equal macro weight.

R99/A | TEST: ALL and18 tasks

TRAIN-fitted SBS and unchanged native/FP64 costs; no per-task checkpoint assembly.

Split / Problem	Top1 (%)	SBS Top1 (%)	mean_cost	SBS cost	Oracle cost	vs_SBS (%)	vs_Oracle (%)	actual regret (%)	TRAIN SBS
ALL	71.3321	66.1802	11.373828	11.385299	11.329932	-0.1986	+0.3845	0.3930	per-problem TRAIN SBS
TSP	47.3111	34.5333	7.970609	7.989621	7.937593	-0.2380	+0.4159	0.4102	T2T
CVRP	47.9556	17.9333	18.010963	18.124983	17.918509	-0.6291	+0.5160	0.6816	RELD_CVRP
ATSP	77.0667	47.2444	1.635246	1.674207	1.629441	-2.3271	+0.3562	0.3566	UNICO_MatPOENet
OVRP	40.7333	33.6444	7.445627	7.455750	7.400487	-0.1358	+0.6099	0.6190	MoSES_RF
VRPB	80.5556	80.4000	9.350241	9.351810	9.315298	-0.0168	+0.3751	0.3774	reld_mt
VRPL	37.2667	32.2667	12.349462	12.356138	12.300958	-0.0540	+0.3943	0.4024	MoSES_RF
VRPTW	81.1111	80.8444	18.689010	18.690545	18.637408	-0.0082	+0.2769	0.2882	reld_mt
OVRPTW	84.0444	83.9333	10.092469	10.093063	10.068135	-0.0059	+0.2417	0.2322	reld_mt
OVRPB	88.1333	88.1333	6.370829	6.370826	6.352225	+0.0000	+0.2929	0.2995	reld_mt
OVRPL	41.6889	34.6000	7.463200	7.472542	7.418033	-0.1250	+0.6089	0.6186	MoSES_RF
VRPBL	79.4222	79.4222	9.395183	9.396379	9.355967	-0.0127	+0.4192	0.4168	reld_mt
VRPBTW	78.8667	78.7556	19.666143	19.665725	19.577666	+0.0021	+0.4519	0.4491	reld_mt
VRPLTW	79.4222	79.2444	18.801405	18.803136	18.745143	-0.0092	+0.3001	0.3170	reld_mt
OVRPBL	89.2667	89.2889	6.375520	6.375741	6.359310	-0.0035	+0.2549	0.2657	reld_mt
OVRPBTW	84.8889	84.8222	10.659032	10.659591	10.624748	-0.0052	+0.3227	0.2956	reld_mt
OVRPLTW	82.9556	82.8667	10.085861	10.086112	10.059001	-0.0025	+0.2670	0.2683	reld_mt
VRPBLTW	79.6889	79.7333	19.705749	19.707068	19.613284	-0.0067	+0.4714	0.4517	reld_mt
OVRPBLTW	83.6000	83.5778	10.662363	10.662149	10.625574	+0.0020	+0.3462	0.3239	reld_mt

vs_SBS=100*(mean_cost/SBS_cost-1); vs_Oracle=100*(mean_cost/Oracle_cost-1).

actual regret=mean_i[100*(selected_i-oracle_i)/oracle_i]; not a ratio of means.

TRAIN81000 / VAL4500 / TEST4500 per problem. All18 problems have equal macro weight.

R99/A | LIB and locked provenance

Current R99 factory, same buffers and selected checkpoint; no solver execution or LIB tuning.

Split / Problem	Top1 (%)	SBS Top1 (%)	mean_cost	SBS cost	Oracle cost	vs_SBS (%)	vs_Oracle (%)	actual regret (%)
TSPLIB	26.5306	10.2041	8.207570	8.240435	8.142167	-0.3988	+0.8033	0.8573
CVRPLIB	64.0000	45.0000	67.240244	67.245364	66.945093	-0.0076	+0.4409	0.5607

SHA256: 37629bc3270a2523c328bf1e7c8d31e501f7bf42edb6e927264168c80dd75d43

Unique selection: strict minimum full VAL macro_vs_sbs_pct with common new81000 TRAIN-fitted SBS; first wins ties

Inference: eval input-only distance messages, A row-softmax or B fixed64 Sinkhorn; FP32 transform then FP16 quantization; outgoing/incoming one/two-hop raw8; Linear32to128 before original four instance layers; R88B physical TSP node0; hanc

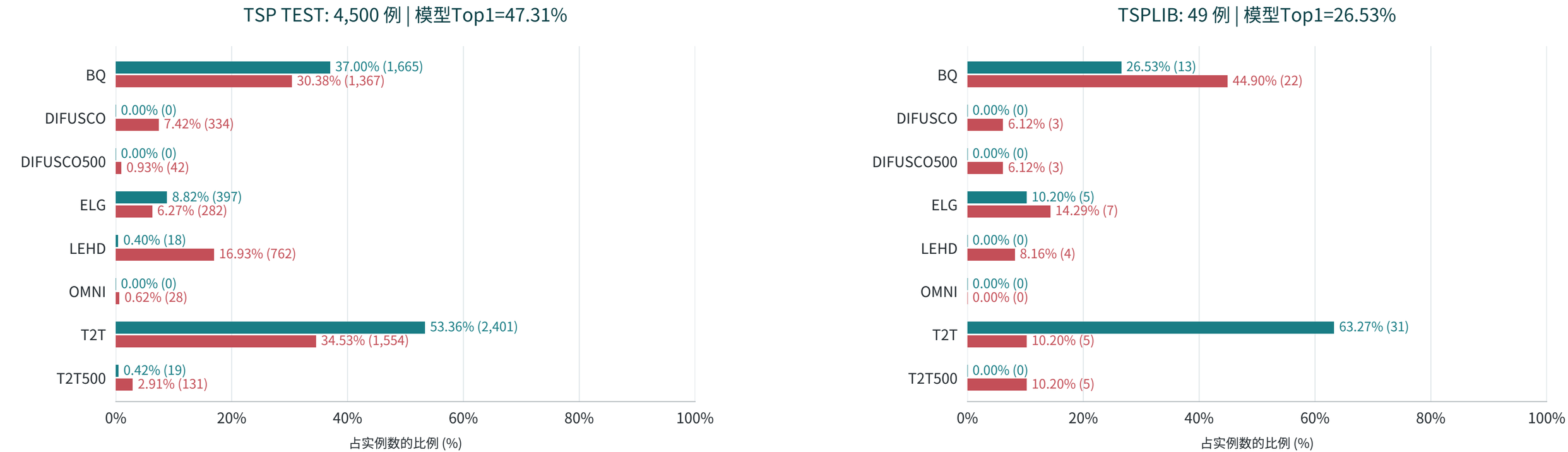
Normalization: new81000 TRAIN only; no TEST/LIB refit.

R99/A | TSP | 方法分布

同一锁定模型: epoch 39 / update 56160 / SHA256 37629bc3270a2523

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

LIB: data/TSPLIB | native/FP64 ind



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

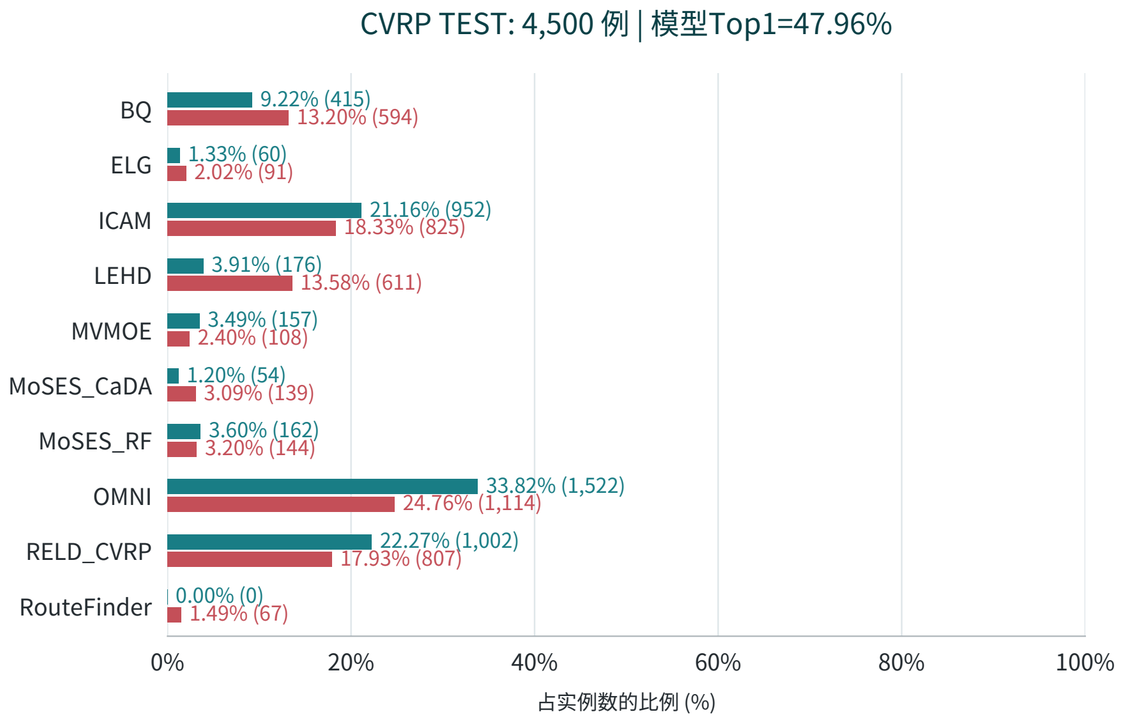
R99/A: stable global solver-ID score ties. Same R99 lock; LIB inference only; no training/reselection/Rxx mixing.

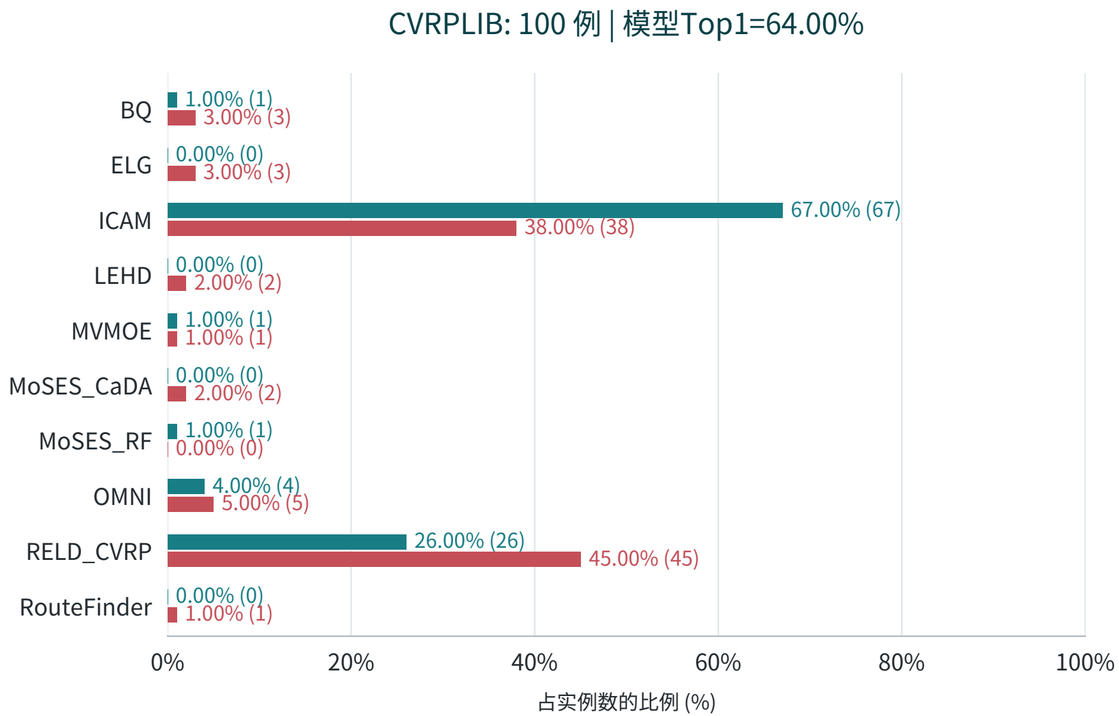
R99/A | CVRP | 方法分布

同一锁定模型: epoch 39 / update 56160 / SHA256 37629bc3270a2523

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

LIB: data/CVRPLIB | native/FP64 ind





标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

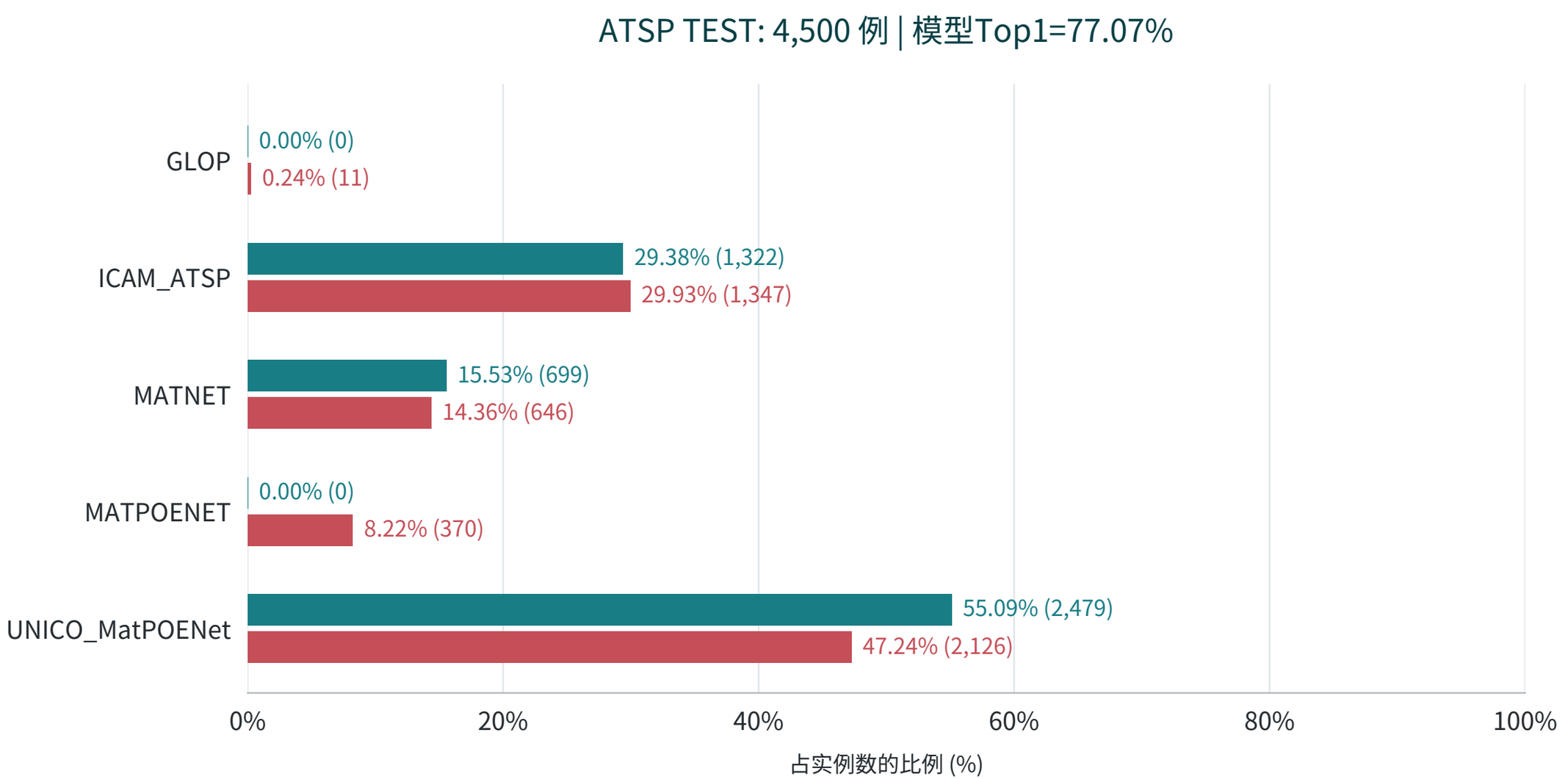
R99/A: stable global solver-ID score ties. Same R99 lock; LIB inference only; no training/reselection/Rxx mixing.

R99/A | ATSP | 方法分布

同一锁定模型: epoch 39 / update 56160 / SHA256 37629bc3270a2523

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

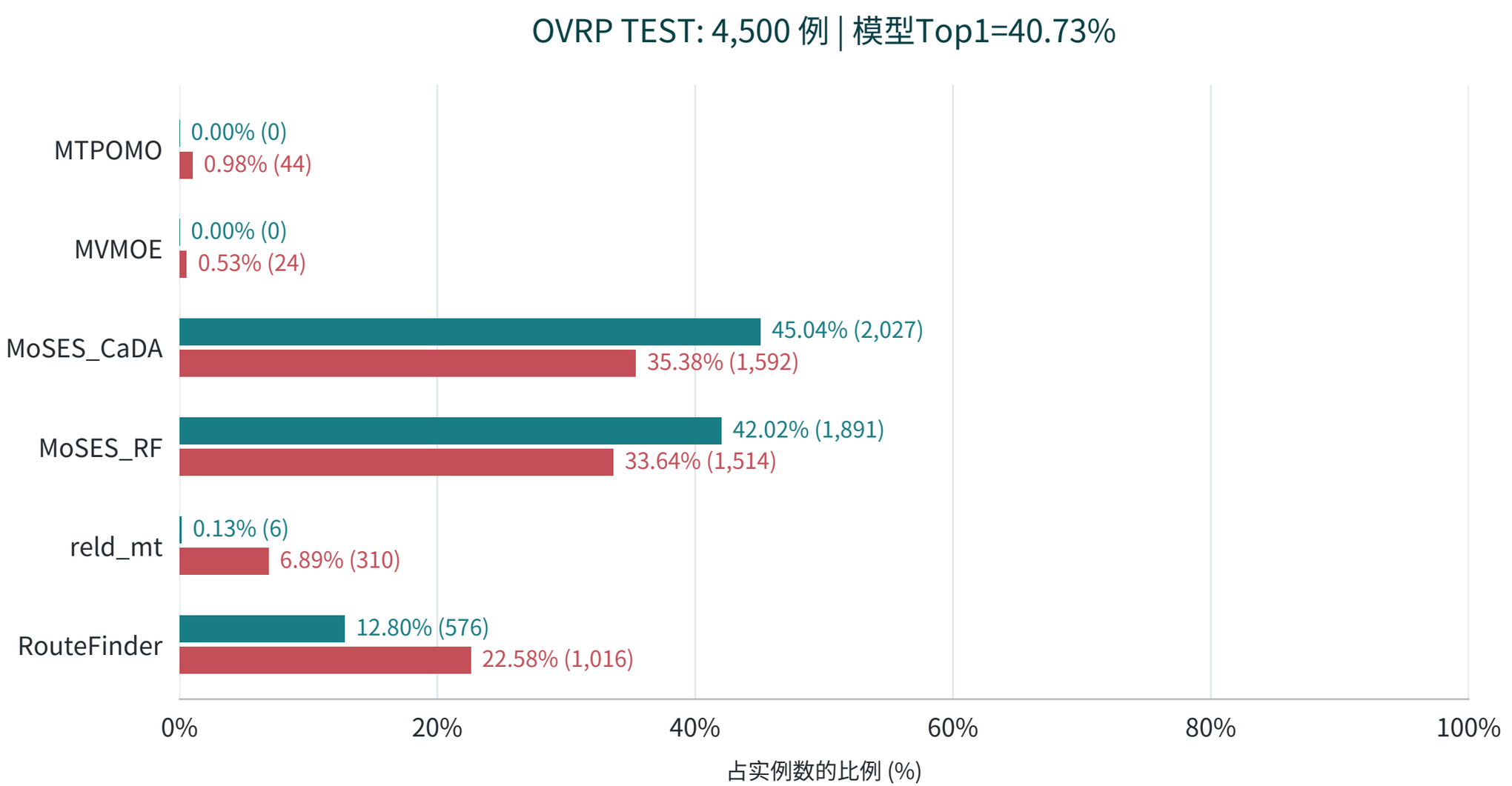
R99/A: stable global solver-ID score ties. Same R99 lock; LIB inference only; no training/reselection/Rxx mixing.

R99/A | OVRP | 方法分布

同一锁定模型: epoch 39 / update 56160 / SHA256 37629bc3270a2523

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

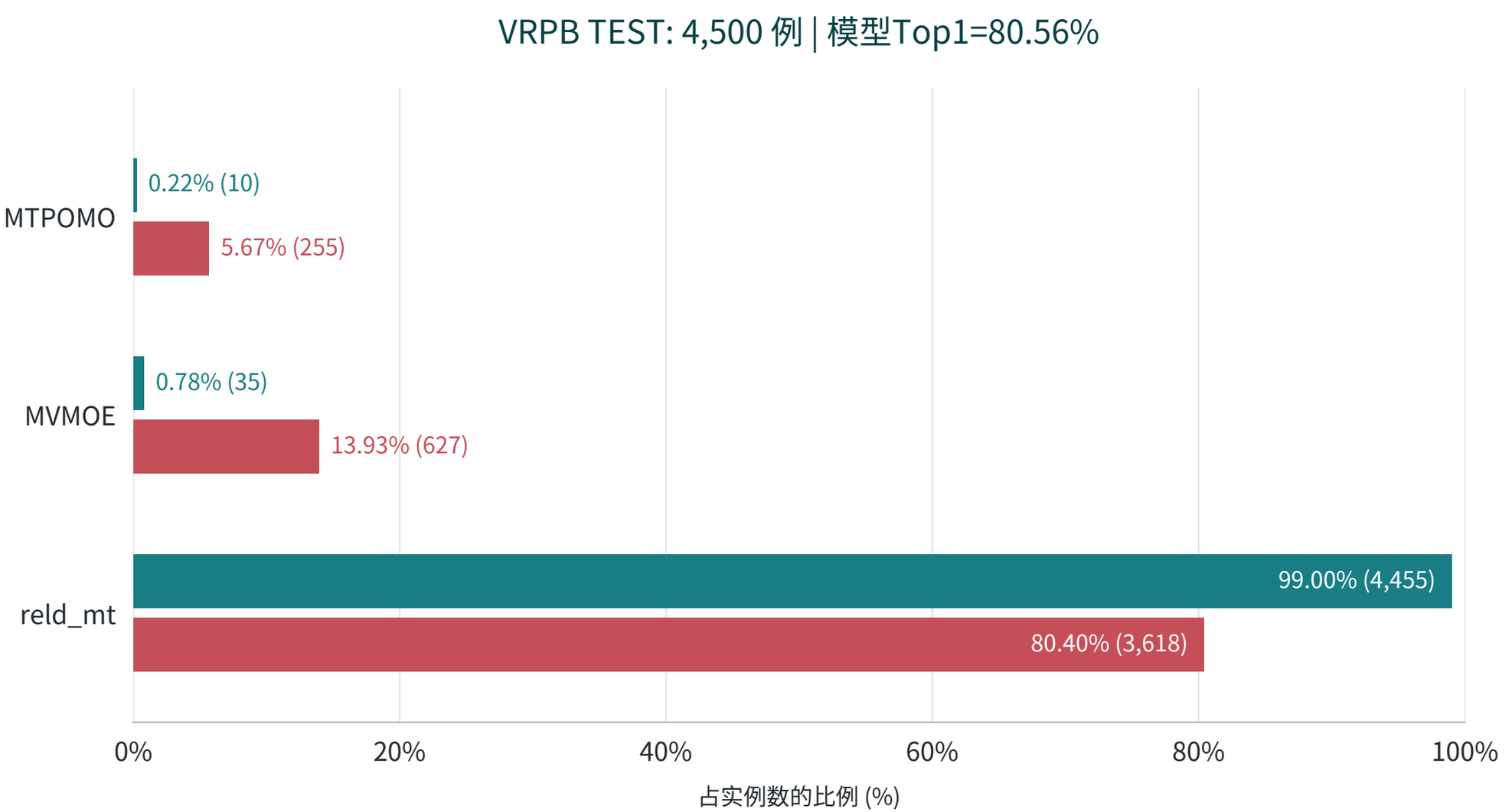
R99/A: stable global solver-ID score ties. Same R99 lock; LIB inference only; no training/reselection/Rxx mixing.

R99/A | VRPB | 方法分布

同一锁定模型: epoch 39 / update 56160 / SHA256 37629bc3270a2523

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

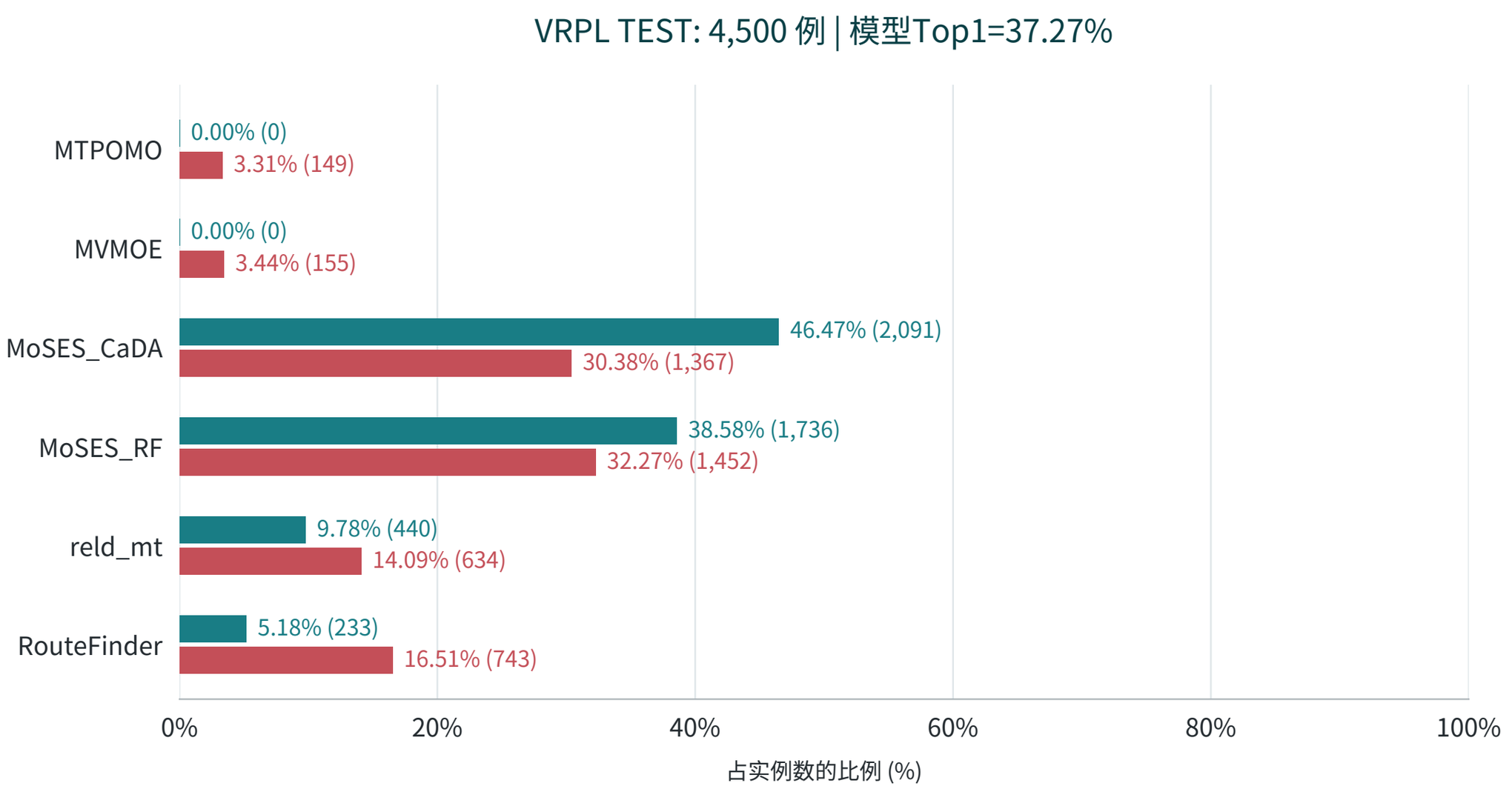
R99/A: stable global solver-ID score ties. Same R99 lock; LIB inference only; no training/reselection/Rxx mixing.

R99/A | VRPL | 方法分布

同一锁定模型: epoch 39 / update 56160 / SHA256 37629bc3270a2523

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

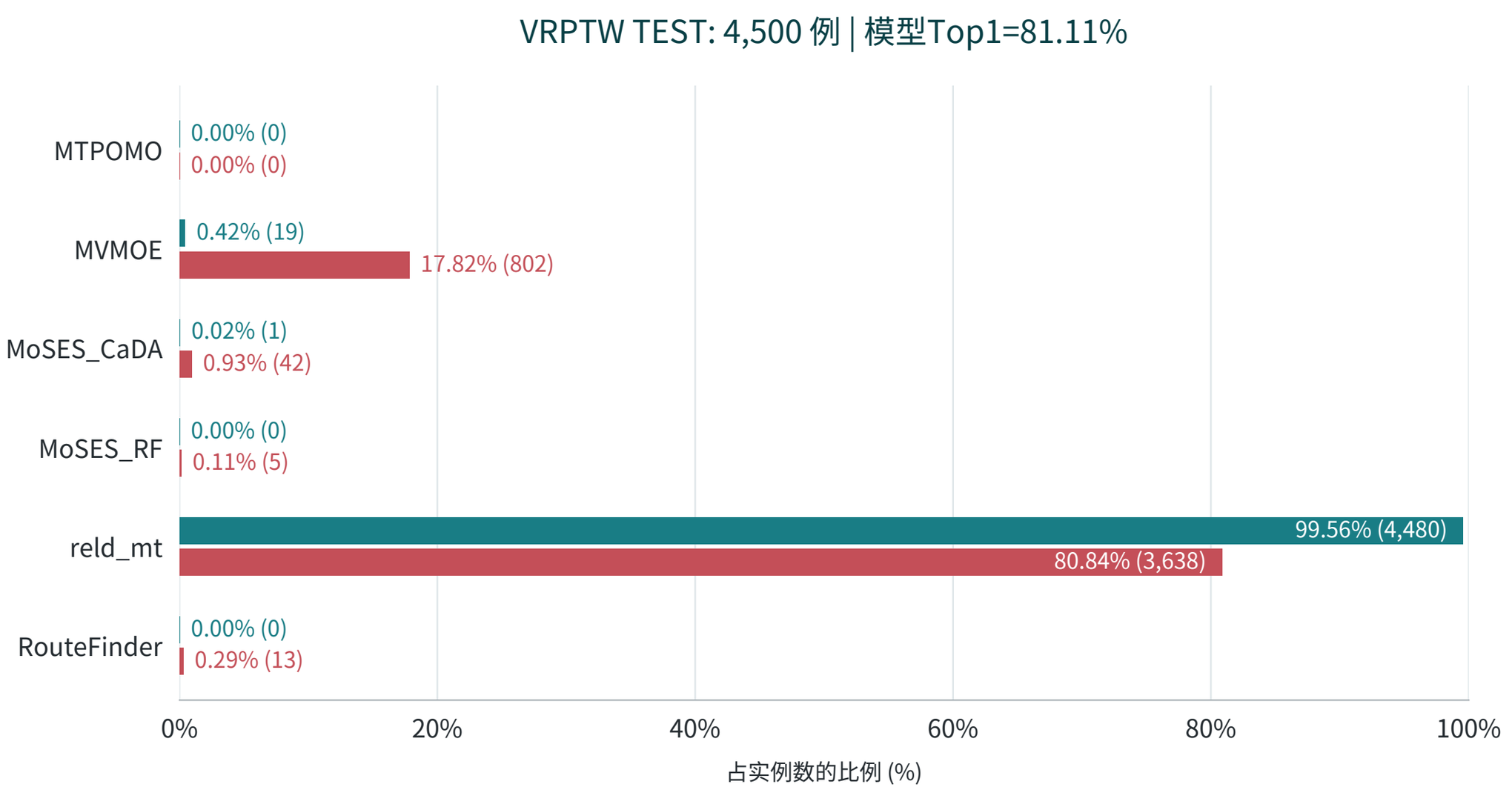
R99/A: stable global solver-ID score ties. Same R99 lock; LIB inference only; no training/reselection/Rxx mixing.

R99/A | VRPTW | 方法分布

同一锁定模型: epoch 39 / update 56160 / SHA256 37629bc3270a2523

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

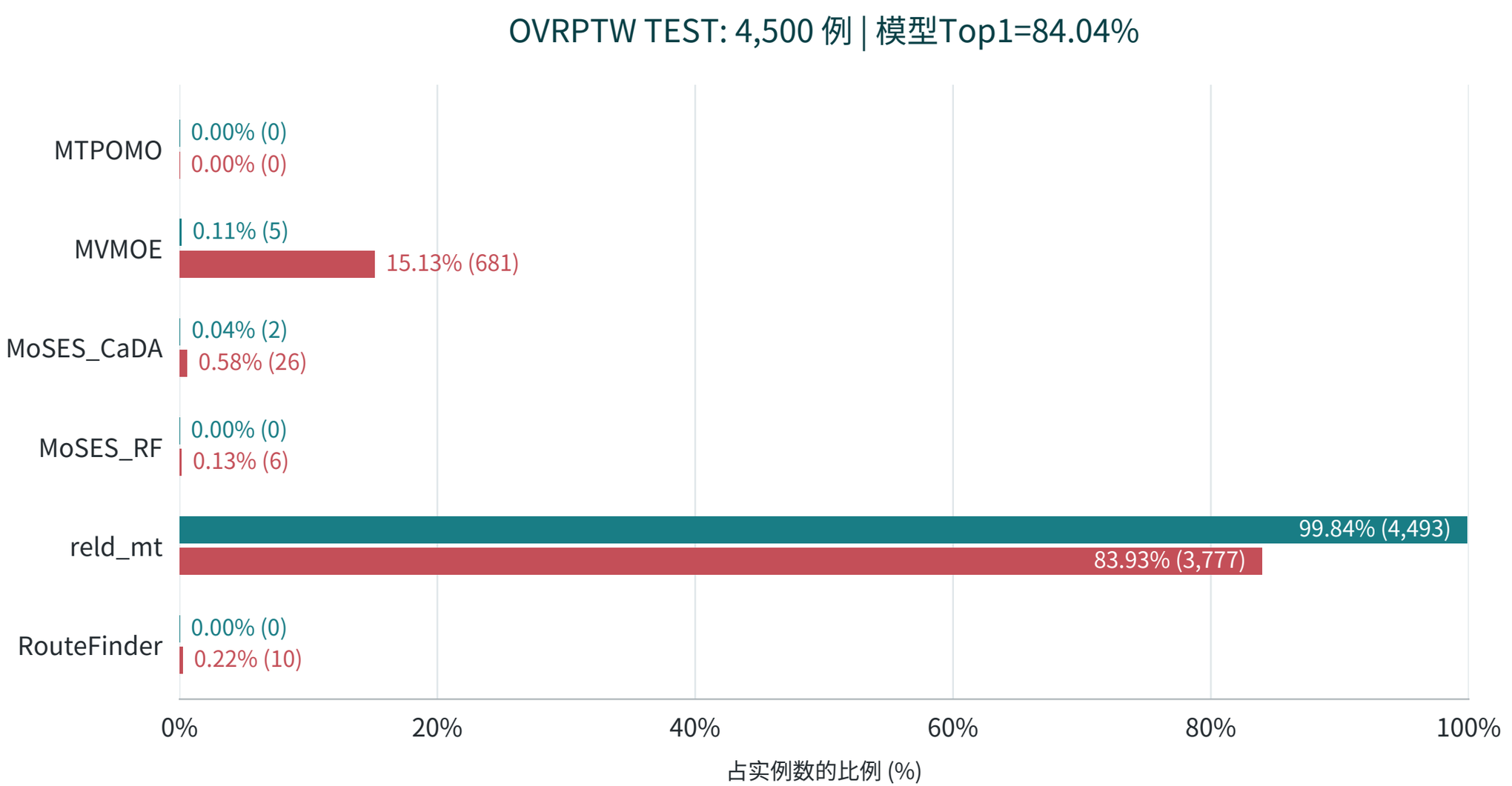
R99/A: stable global solver-ID score ties. Same R99 lock; LIB inference only; no training/reselection/Rxx mixing.

R99/A | OVRPTW | 方法分布

同一锁定模型: epoch 39 / update 56160 / SHA256 37629bc3270a2523

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

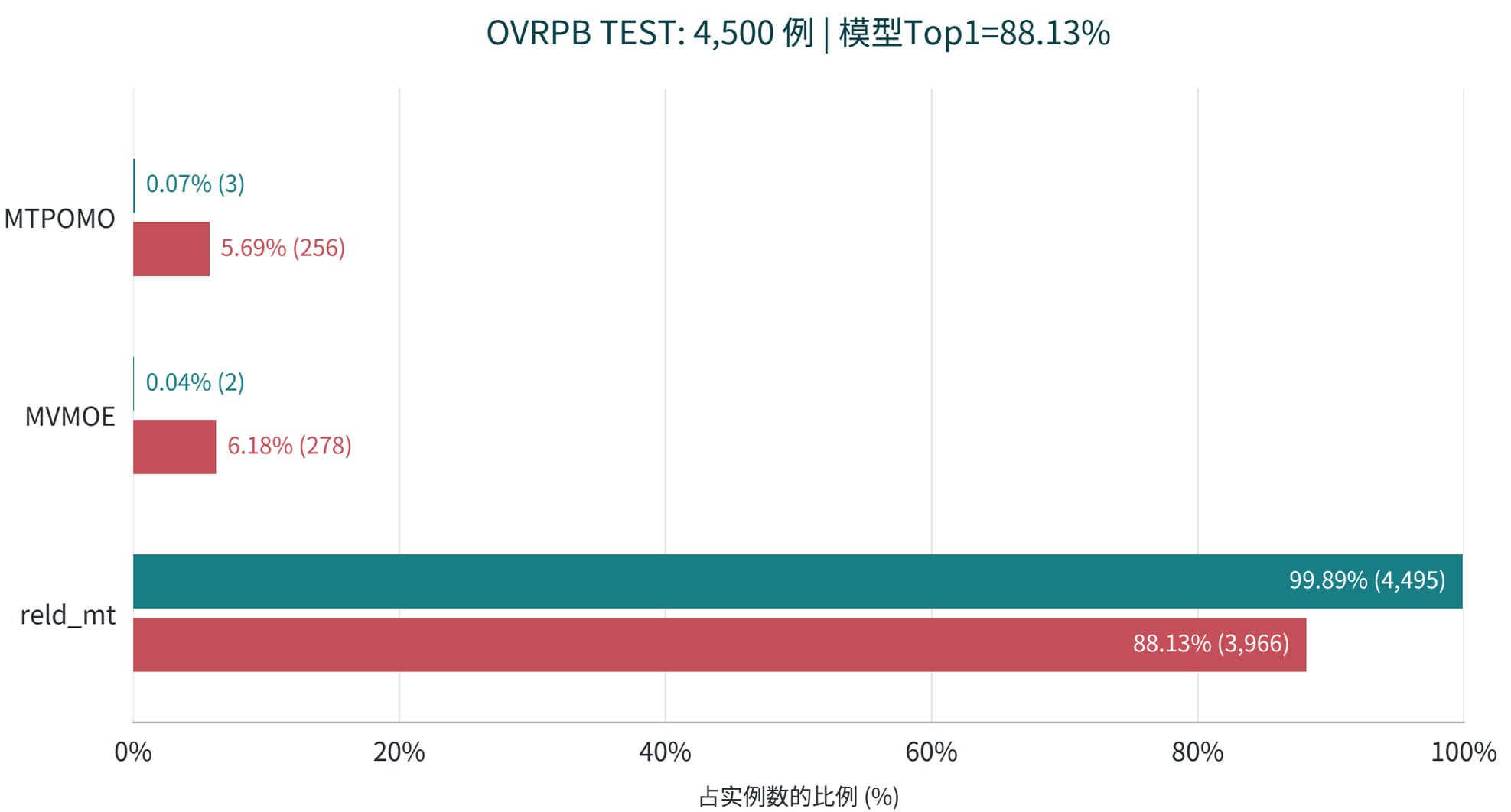
R99/A: stable global solver-ID score ties. Same R99 lock; LIB inference only; no training/reselection/Rxx mixing.

R99/A | OVRPB | 方法分布

同一锁定模型: epoch 39 / update 56160 / SHA256 37629bc3270a2523

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

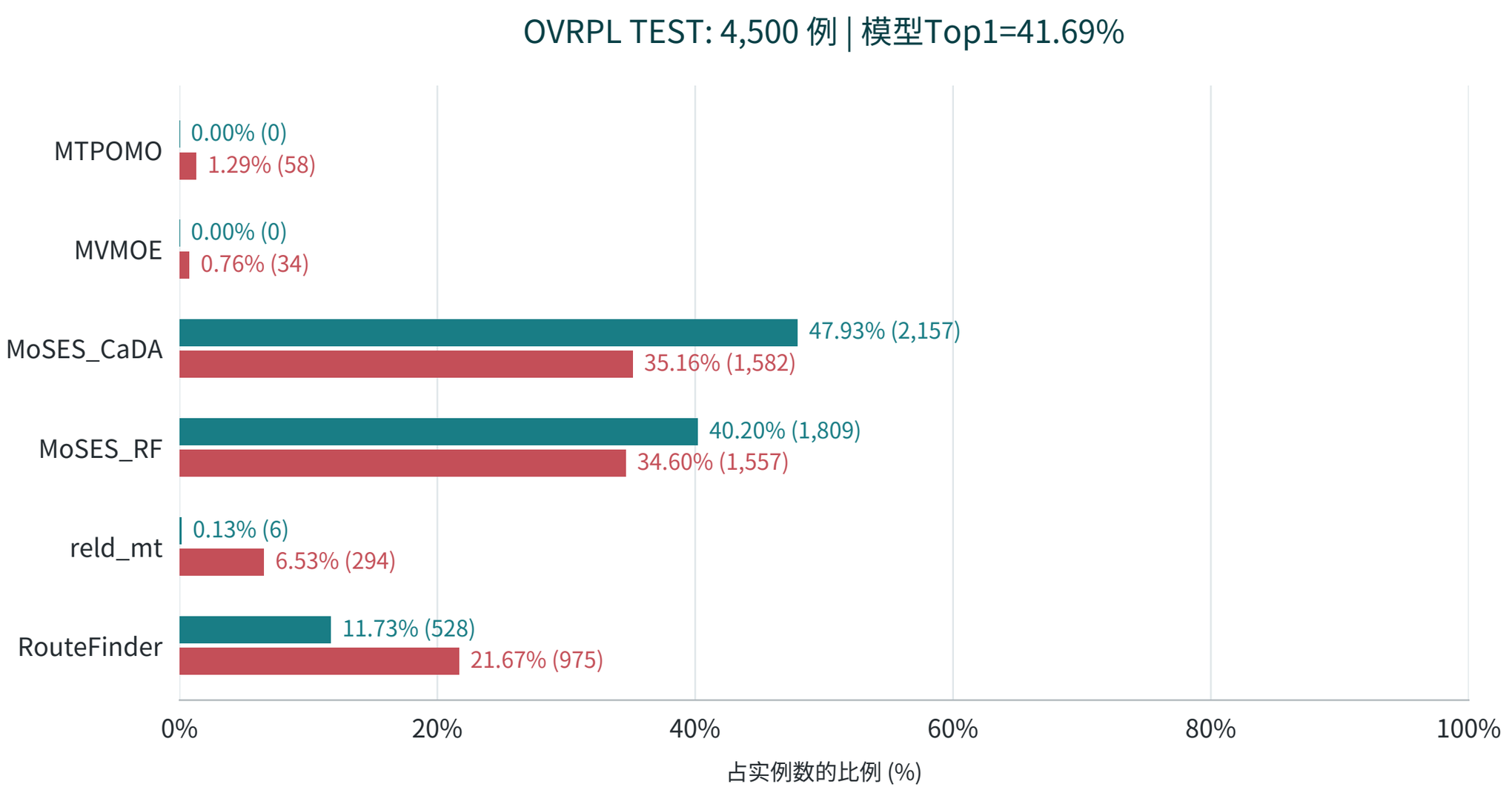
R99/A: stable global solver-ID score ties. Same R99 lock; LIB inference only; no training/reselection/Rxx mixing.

R99/A | OVRPL | 方法分布

同一锁定模型: epoch 39 / update 56160 / SHA256 37629bc3270a2523

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

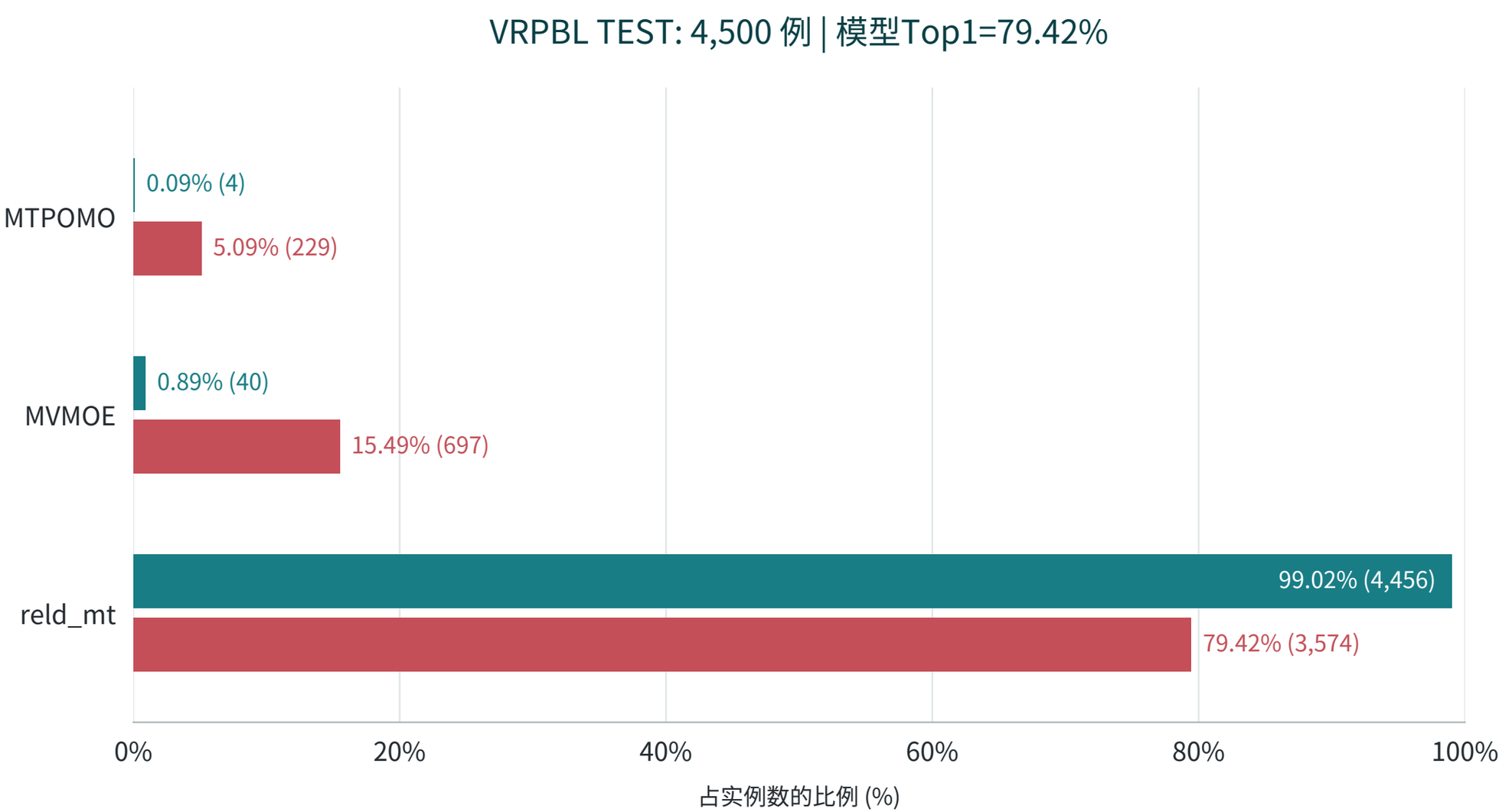
R99/A: stable global solver-ID score ties. Same R99 lock; LIB inference only; no training/reselection/Rxx mixing.

R99/A | VRPBL | 方法分布

同一锁定模型: epoch 39 / update 56160 / SHA256 37629bc3270a2523

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

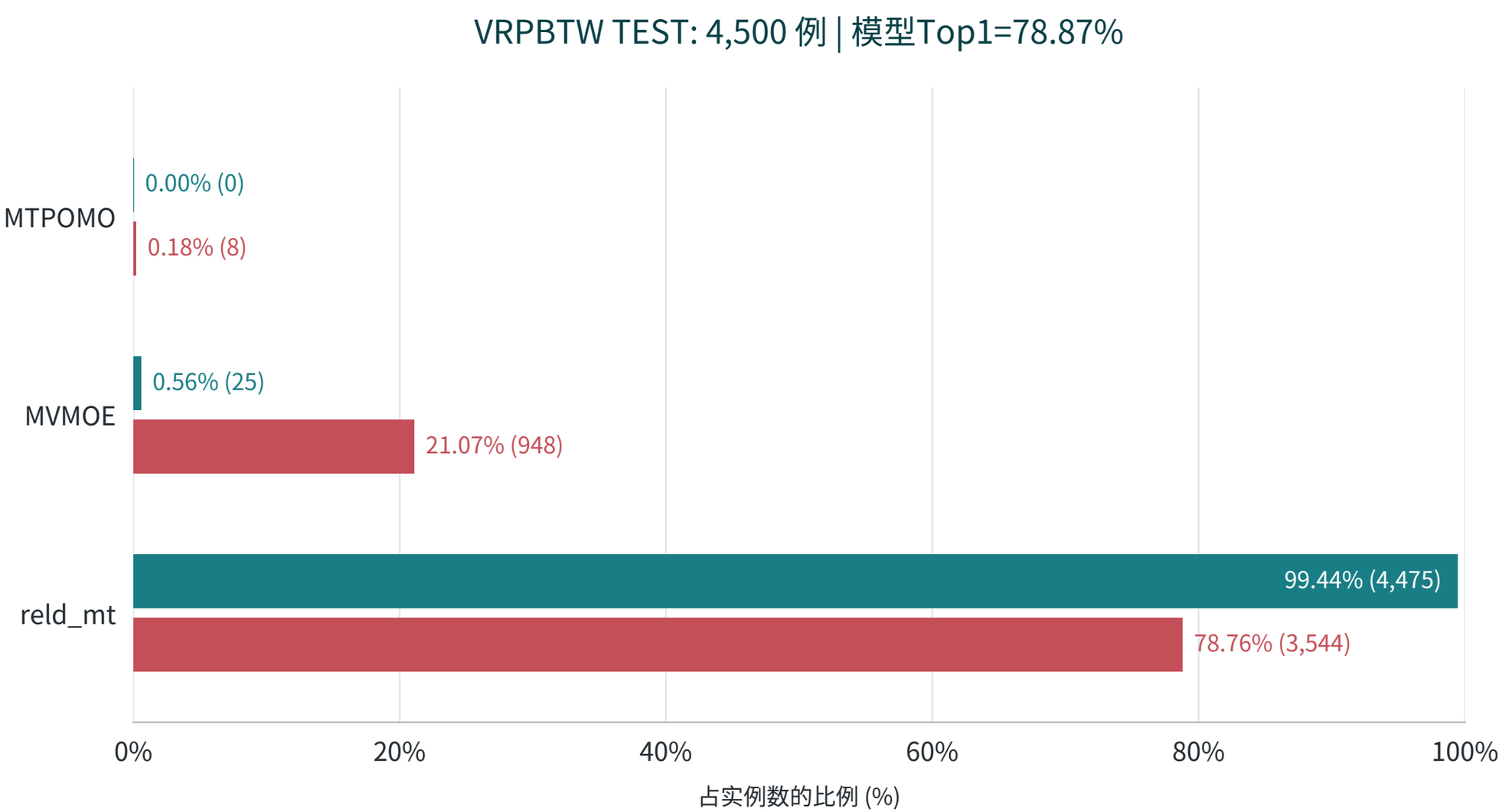
R99/A: stable global solver-ID score ties. Same R99 lock; LIB inference only; no training/reselection/Rxx mixing.

R99/A | VRPBTW | 方法分布

同一锁定模型: epoch 39 / update 56160 / SHA256 37629bc3270a2523

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

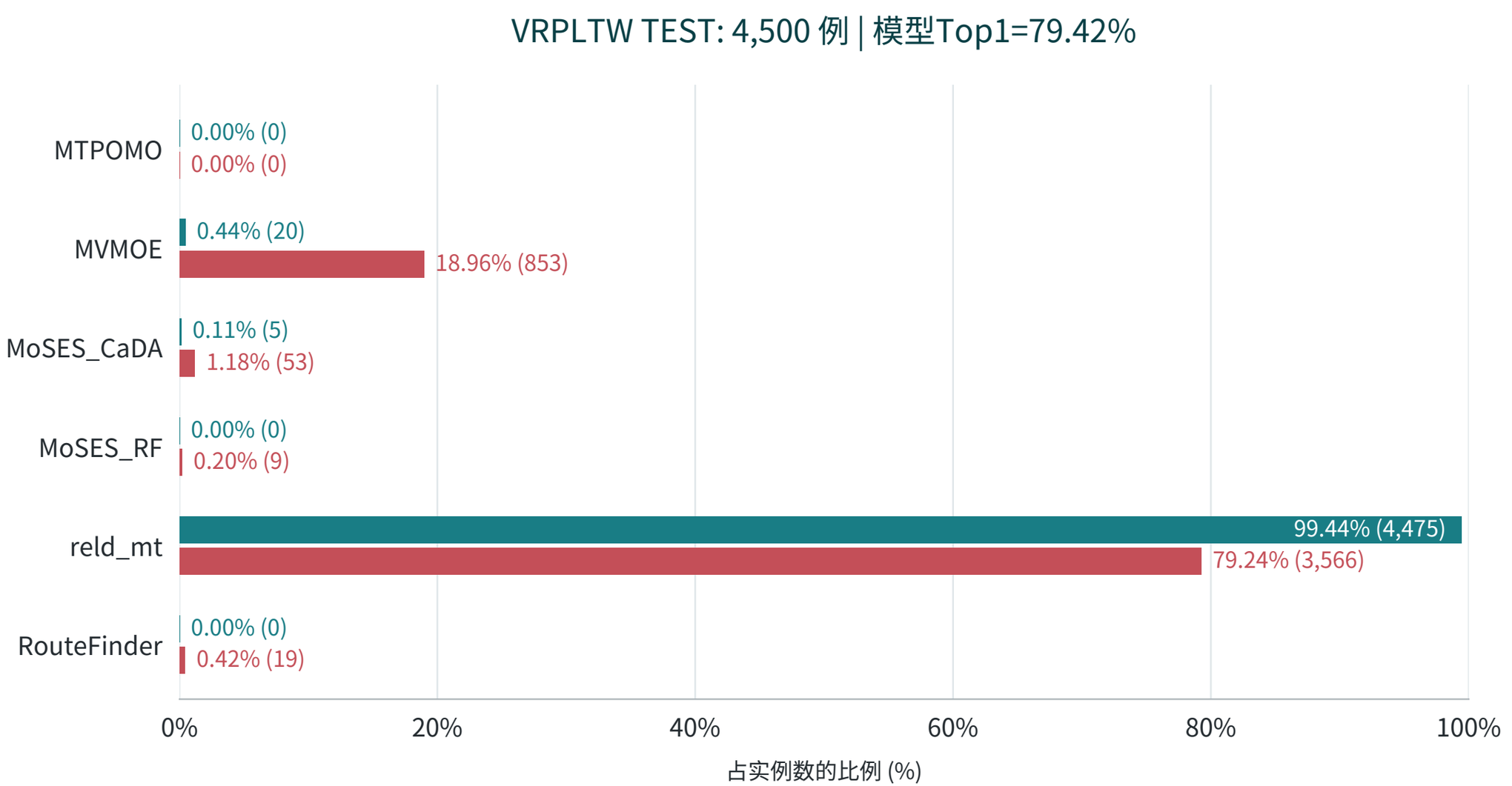
R99/A: stable global solver-ID score ties. Same R99 lock; LIB inference only; no training/reselection/Rxx mixing.

R99/A | VRPLTW | 方法分布

同一锁定模型: epoch 39 / update 56160 / SHA256 37629bc3270a2523

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

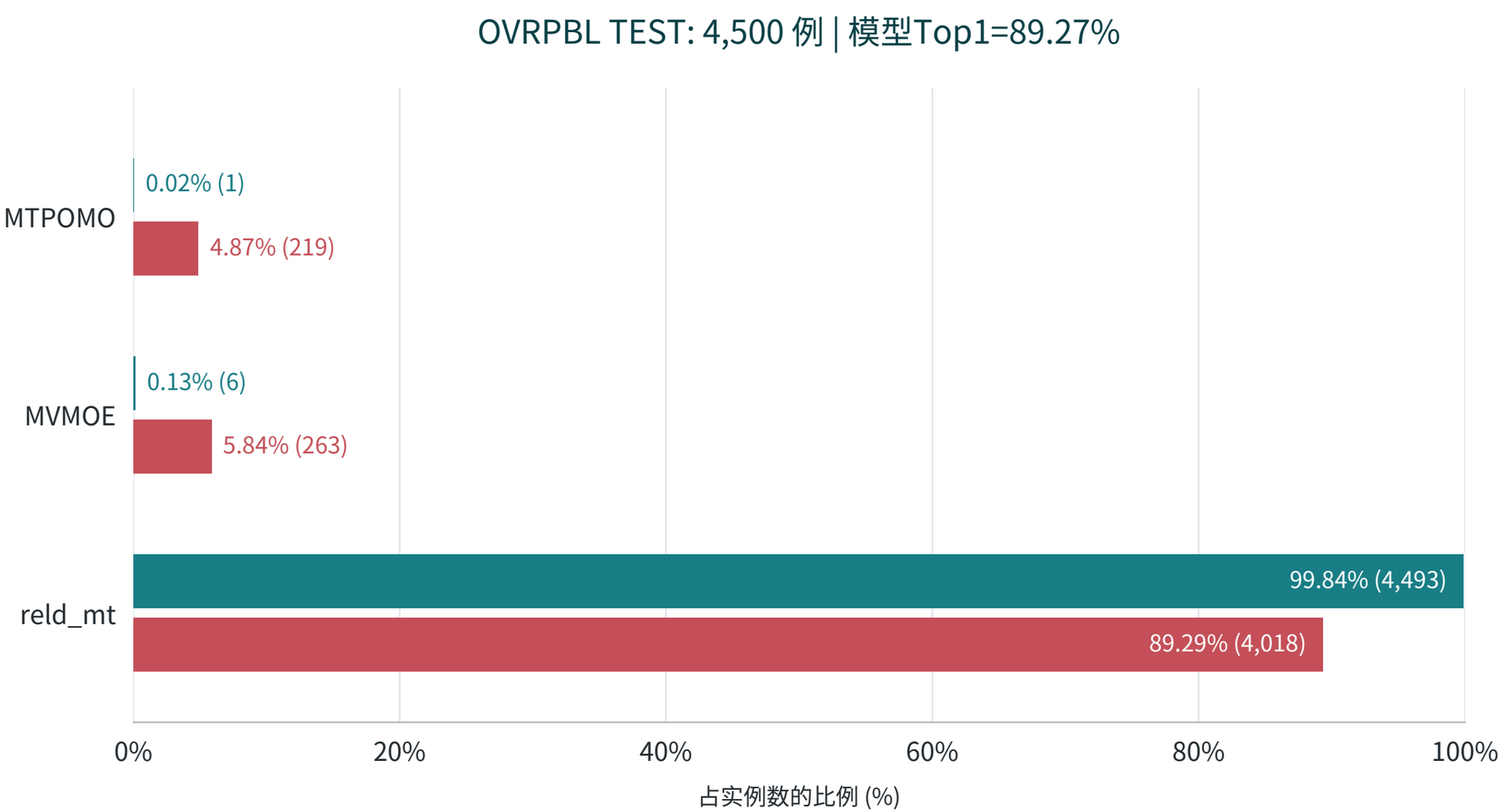
R99/A: stable global solver-ID score ties. Same R99 lock; LIB inference only; no training/reselection/Rxx mixing.

R99/A | OVRPBL | 方法分布

同一锁定模型: epoch 39 / update 56160 / SHA256 37629bc3270a2523

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

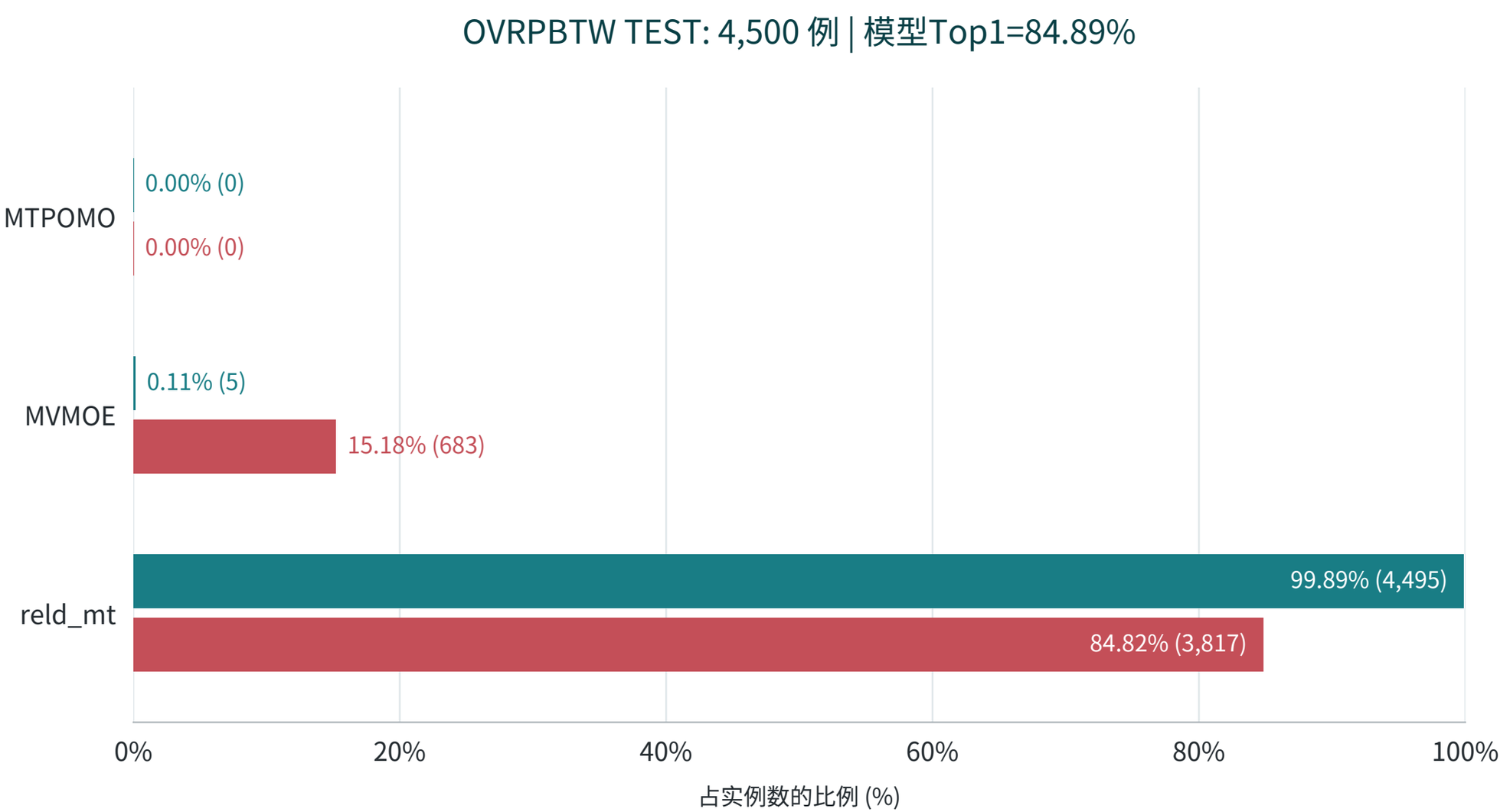
R99/A: stable global solver-ID score ties. Same R99 lock; LIB inference only; no training/reselection/Rxx mixing.

R99/A | OVRPBTW | 方法分布

同一锁定模型: epoch 39 / update 56160 / SHA256 37629bc3270a2523

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

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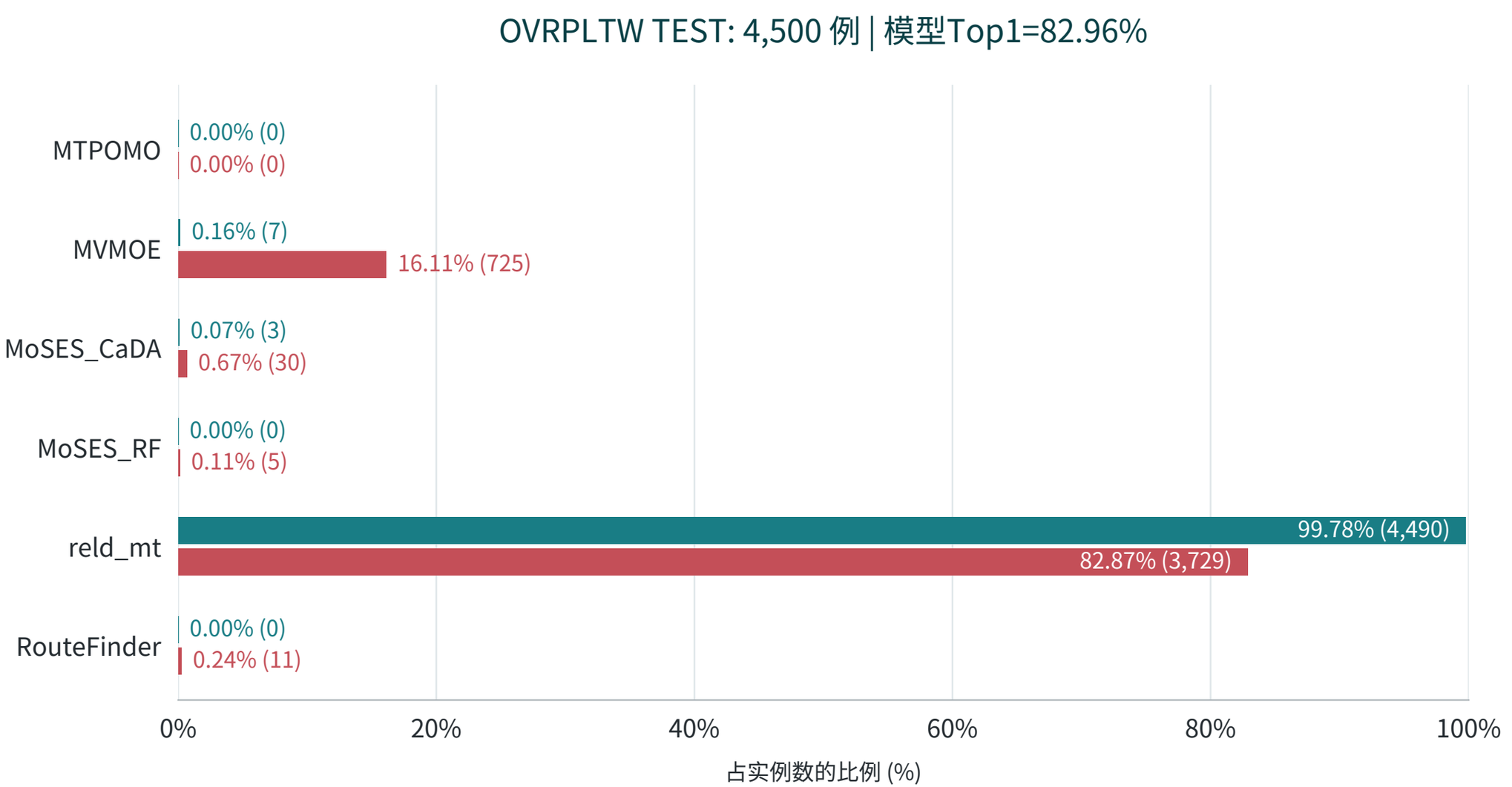
R99/A: stable global solver-ID score ties. Same R99 lock; LIB inference only; no training/reselection/Rxx mixing.

R99/A | OVRPLTW | 方法分布

同一锁定模型: epoch 39 / update 56160 / SHA256 37629bc3270a2523

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



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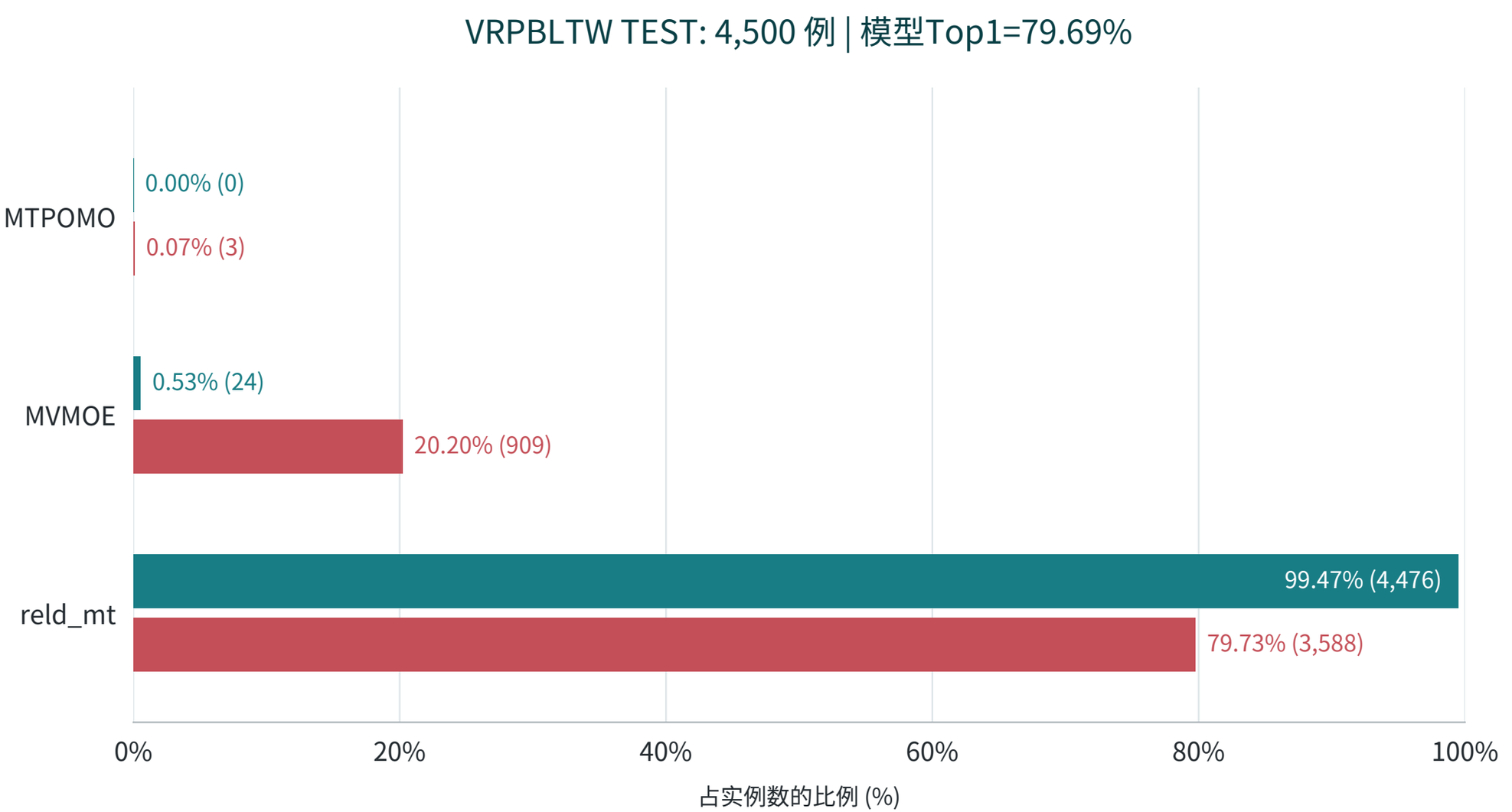
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R99/A | VRPBLTW | 方法分布

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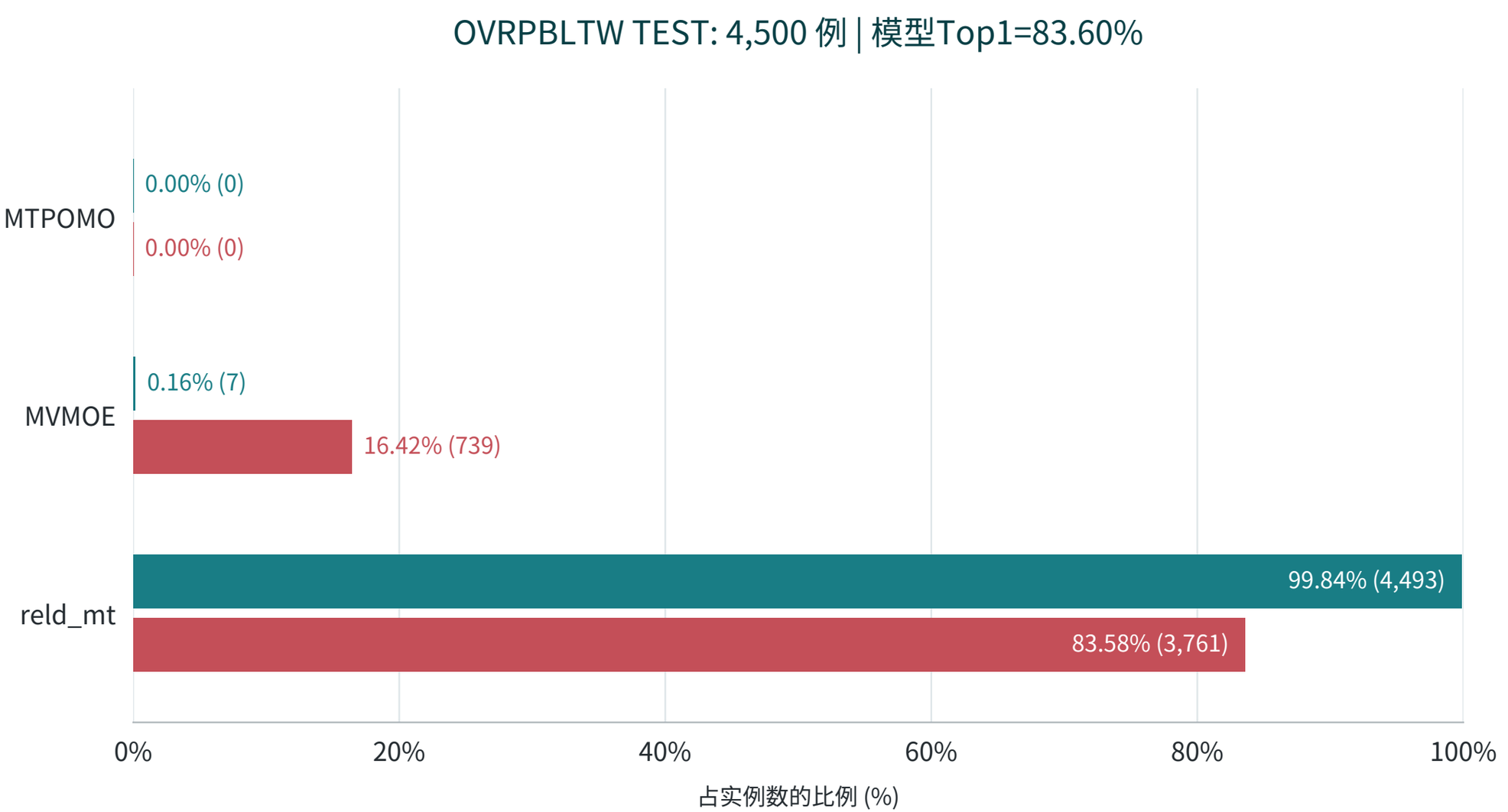
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